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Batch 15

Algorithm

1. Read the `size` of the array.
2. Declare the variable-length array `arr[size]`.
3. Read all elements into `arr`.
4. Pass the address of the array to `max_subarray_sum()`.
5. Initialize:
   - `current_sum = arr[0]`
   - `max_sum = arr[0]`
6. Starting from the second element:
   - Compare the current element with `current_sum + current element`.
   - If the current element is larger, start a new subarray from it.
   - Otherwise, extend the existing subarray.
7. If `current_sum > max_sum`, update `max_sum`.
8. After processing all elements, display `max_sum`.

Initializing both sums with the **first array element** is important because it ensures that an all-negative array returns the **least negative element** rather than `0`.

Test Case Table

Test	Input Array	Expected Output
1	-2 1 -3 4 -1 2 1 -5 4	Maximum subarray sum: 6
2	-8 -3 -6 -2 -5 -4	Maximum subarray sum: -2
3	5 -2 3 4	Maximum subarray sum: 10
4	1 2 3 4 5	Maximum subarray sum: 15
5	-5	Maximum subarray sum: -5
6	-4 -2 -7 -1 -8	Maximum subarray sum: -1
7	0 0 0	

C program

```
#include <stdio.h>

void max_subarray_sum(int *arr, int size)
{
    int current_sum, max_sum;
    int i;

    /* Initialize with first element so that
       all-negative arrays are handled correctly */
    current_sum = *arr;
    max_sum = *arr;

    for (i = 1; i < size; i++)
    {
        /* Decide whether to start a new subarray
           or continue the existing subarray */
        if (*(arr + i) > current_sum + *(arr + i))
        {
            current_sum = *(arr + i);
        }
        else
        {
            current_sum = current_sum + *(arr + i);
        }

        /* Update maximum sum */
        if (current_sum > max_sum)
        {
            max_sum = current_sum;
        }
    }

    printf("Maximum subarray sum: %d\n", max_sum);
}

int main()
```

```
{
    int size, i;

    printf("Enter the size of the array: ");
    scanf("%d", &size);

    if (size <= 0)
    {
        printf("Invalid size\n");
        return 0;
    }

    /* Variable-length array */
    int arr[size];

    printf("Enter %d array elements:\n", size);

    for (i = 0; i < size; i++)
    {
        scanf("%d", &arr[i]);
    }

    /* Pass array using its address */
    max_subarray_sum(arr, size);

    return 0;
}
```

Execution

